11. *(a)* Esters are formed when carboxylic acids are heated with alcohols in the presence of concentrated sulfuric acid. This is a reversible reaction that is carried out by heating the reagents under reflux.

(i) Draw a labelled diagram of the apparatus you would use to carry out a reaction under reflux. [3]

(ii) Explain how the apparatus that you have drawn in part (i) results in the process of reflux. [1]

....

....

(iii) Why is it necessary to use a reflux technique in this type of reaction? [1]

....

....

(iv) State the function of the concentrated sulfuric acid in this reaction. [1]

....

Examiner only

(b) The equation shows the reaction between ethanoic acid and methanol.

$$CH_3COOH + CH_3OH \rightleftharpoons CH_3COOCH_3 + H_2O$$

When 3.00 g of ethanoic acid was heated under reflux with 1.28 g of methanol in the presence of concentrated sulfuric acid for 10 minutes it was found that 1.18 g of methyl ethanoate was formed.

(i) Suggest how the methyl ethanoate could be separated from the mixture formed in the reaction vessel. [1]

....

(ii) Calculate the percentage of theoretical yield of methyl ethanoate obtained. Show your working. [3]

Percentage yield = ... %

(iii) Suggest **one** change that could be made to this preparation to improve the percentage yield of methyl ethanoate. Explain your suggestion. [2]

....

....

Examiner only

(c) Two reactions of organic compound **A** are shown below.

$$\mathbf{B} \xleftarrow{\text{conc } H_2SO_4} \mathbf{A} \xrightarrow{Cr_2O_7^{2-}/H^+} \mathbf{C}$$

Compound **B** is a straight-chain hydrocarbon with the formula C_4H_8.

(i) What type of reaction is taking place when compound **A** is heated with concentrated sulfuric acid to form compound **B**? [1]

....

(ii) Draw the displayed formulae of **two** possible isomers of **A**. [2]

(iii) What colour **change** is seen when compound **A** reacts with the $Cr_2O_7^{2-}/H^+$ solution? [1]

....

(iv) What type of reaction is taking place when compound **A** is heated with $Cr_2O_7^{2-}/H^+$ to form compound **C**? [1]

....

(v) Draw a displayed formula of compound **C**. [1]

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Examiner only

11. *(a)* Ethanol can be prepared by the fermentation of glucose, $C_6H_{12}O_6$. The equation for this reaction is as follows.

$$C_6H_{12}O_6(aq) \longrightarrow 2C_2H_5OH(aq) + 2CO_2(g)$$

(i) A student wanted to prepare ethanol and was given the following instructions.

- Mix glucose and water.
- Put the mixture into a stoppered bottle.
- Leave the bottle for several weeks.
- Separate off the ethanol.

Suggest how the student should modify these instructions in order to obtain a good yield of pure ethanol. [3]

....

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....

(ii) Calculate the atom economy of this process as a method of preparing ethanol. [2]

Atom economy = ... %

(iii) Ethanol can also be made by the hydration of ethene.

Describe **one** environmental consideration in choosing between fermentation and hydration as methods for ethanol production. [2]

....

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Examiner only

(b) Alcohols are classified as being primary, secondary or tertiary.

(i) Explain the meaning of *primary* as applied to alcohols. [1]

....

....

(ii) Give the structural formula and name of a tertiary alcohol that contains five carbon atoms. [2]

Name ....

(iii) Describe **one** test that can be used to investigate whether an alcohol is secondary or tertiary. Include the reagents used and the observations for both types of alcohol. [3]

....

....

....

....

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END OF PAPER

Surname	Centre Number	Candidate Number
Other Names		2

GCE AS/A LEVEL

2410U20-1

S18-2410U20-1

CHEMISTRY – AS unit 2

Energy, Rate and Chemistry of Carbon Compounds

FRIDAY, 25 MAY 2018 – MORNING

1 hour 30 minutes

	For Examiner's use only		
	Question	**Maximum Mark**	**Mark Awarded**
Section A	**1. to 7.**	**10**	
Section B	**8.**	**11**	
	9.	**12**	
	10.	**16**	
	11.	**12**	
	12.	**13**	
	13.	**6**	
	Total	**80**	

2410U201
01

ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:
- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions in the spaces provided.
Section B Answer **all** questions in the spaces provided.

Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

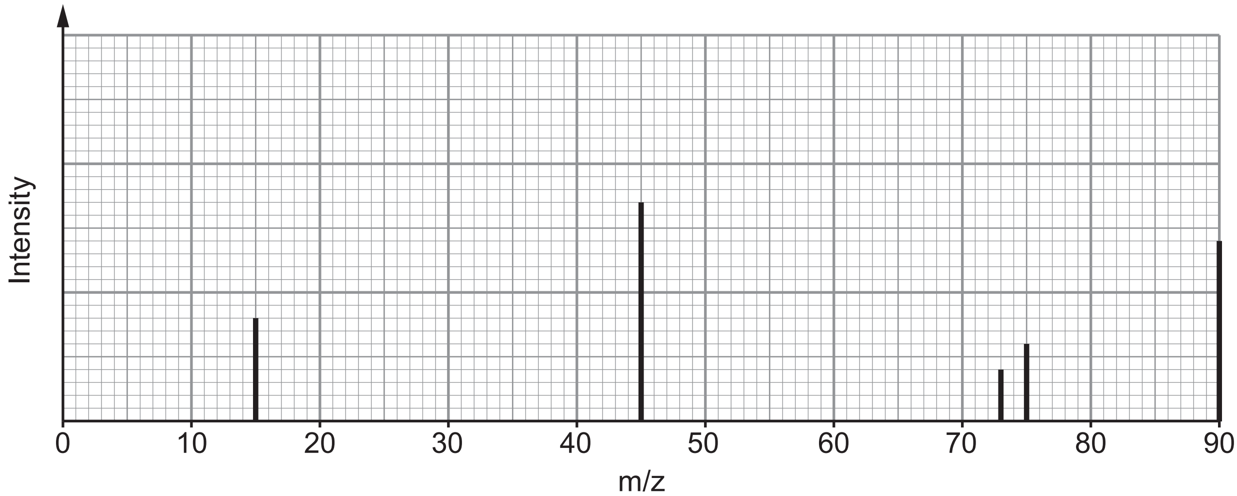
Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

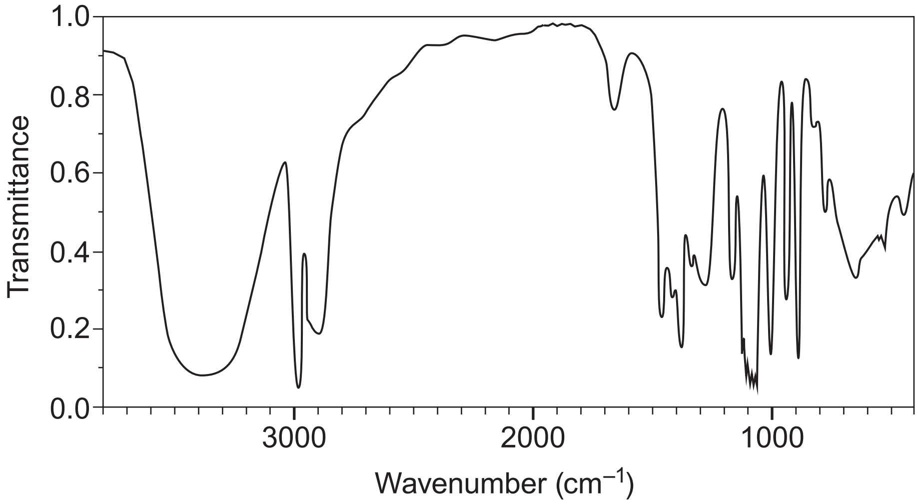
The assessment of the quality of extended response (QER) will take place in **Q.10**(*a*)(i).

If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

11. Compound **X** contains only carbon, hydrogen and oxygen. On analysis it was found to contain 53.3 % carbon and 11.1 % hydrogen by mass.

A simplified form of the mass spectrum and the infrared absorption spectrum for **X** are shown.





The low resolution ^1H NMR spectrum of **X** has three peaks.

When **X** is warmed with excess acidified potassium dichromate(VI) there is a colour change. The organic product of this reaction does **not** react with aqueous sodium carbonate.

Examiner only

(a) Use **all** the data given to find the structure of compound **X**. Explain what information can be found from each piece of data. [10]

Structure of **X**

Examiner only

(b) (i) State the type of reaction that occurs when **X** is warmed with acidified potassium dichromate(VI). [1]

....

(ii) Draw the structure of the organic product formed when **X** reacts with acidified potassium dichromate(VI). [1]

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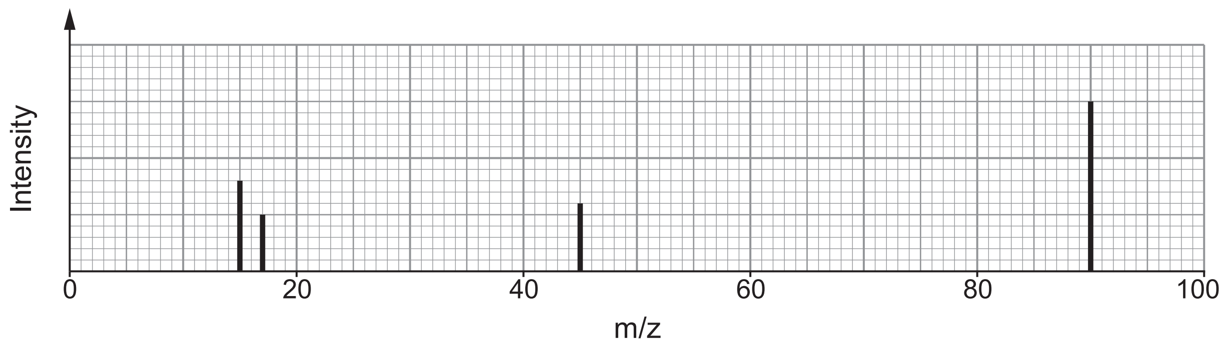
11. *(a)* Compound **X** contains only carbon, hydrogen and oxygen.

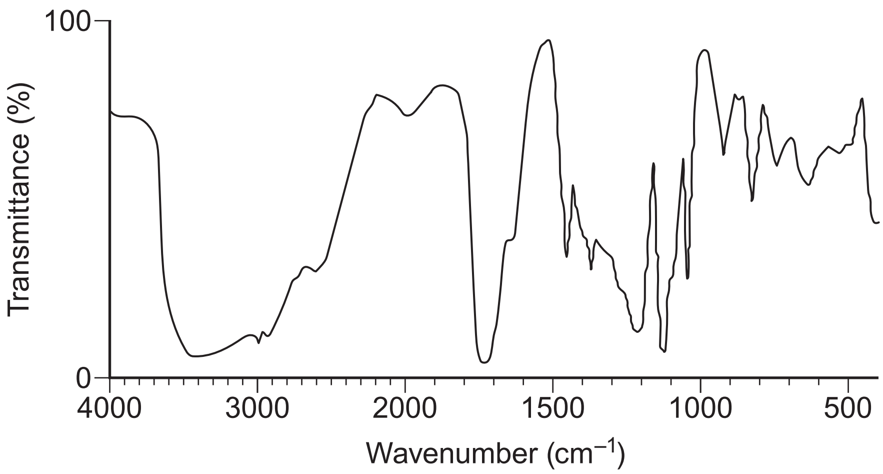
On analysis **X** was found to contain 40.0 % carbon and 6.67 % hydrogen by mass.

A simplified form of the mass spectrum, the IR spectrum and the low resolution ^{1}H NMR spectrum of **X** are shown.

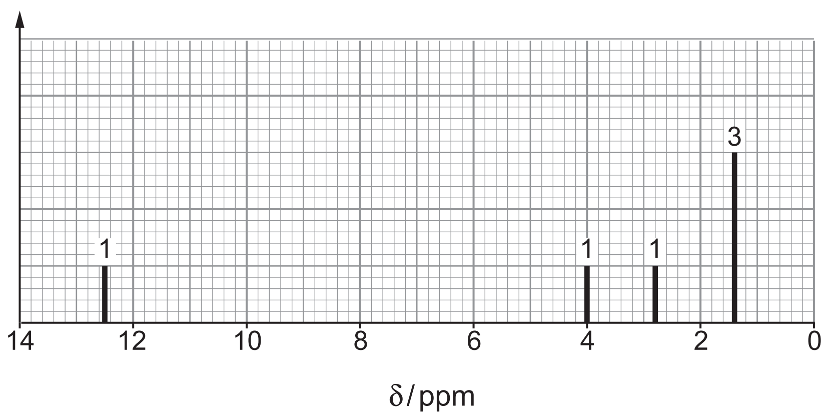
Mass spectrum



IR spectrum



^{1}H NMR spectrum



The numbers above the peaks show the relative areas of the peaks.

Examiner only

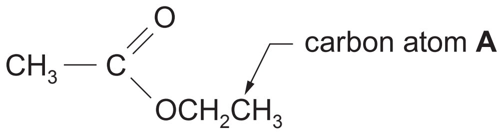
Use this information to identify compound **X**.

You must use information from **all** the sources given and explain how you used it. [10]

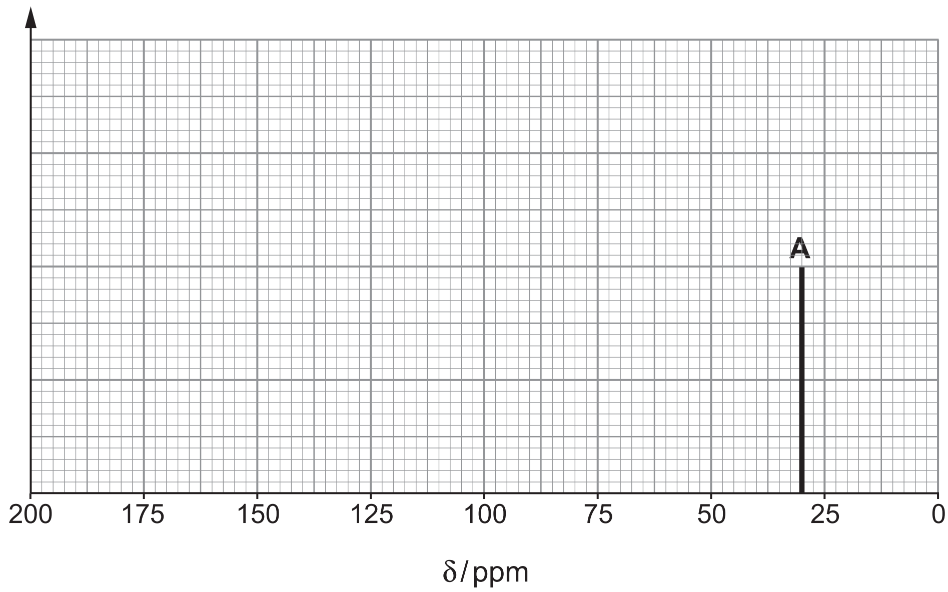
Examiner only

(b) Information about the structure of organic compounds can also be found using a ^{13}C NMR spectrum.

(i) On the axes below sketch the ^{13}C NMR spectrum that you would expect to obtain from ethyl ethanoate.



Label the diagram to show the species causing each peak.
The peak caused by carbon atom **A** is already included. [3]



(ii) A student said that valuable information about the nature of a sample being analysed could be obtained from the peak heights in both the ^{13}C and ^{1}H NMR spectra.

Comment on whether the student is correct. Give a reason for your answer. [1]

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END OF PAPER

Surname	Centre Number	Candidate Number
First name(s)		2

GCE AS/A LEVEL

2410U20-1

Z22-2410U20-1

FRIDAY, 27 MAY 2022 – AFTERNOON

CHEMISTRY – AS unit 2

Energy, Rate and Chemistry of Carbon Compounds

1 hour 30 minutes

	For Examiner's use only		
	Question	**Maximum Mark**	**Mark Awarded**
Section A	**1. to 5.**	**10**	
Section B	**6.**	**15**	
	7.	**17**	
	8.	**10**	
	9.	**12**	
	10.	**16**	
	Total	**80**	

2410U201
01

ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:
- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions.
Section B Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

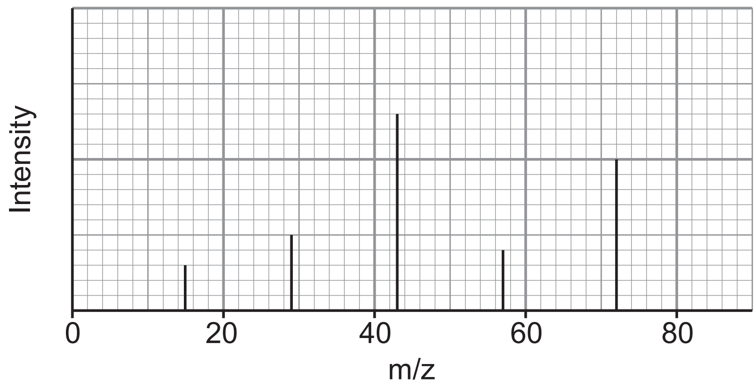
The assessment of the quality of extended response (QER) will take place in **Q9**(c).

JUN222410U20101

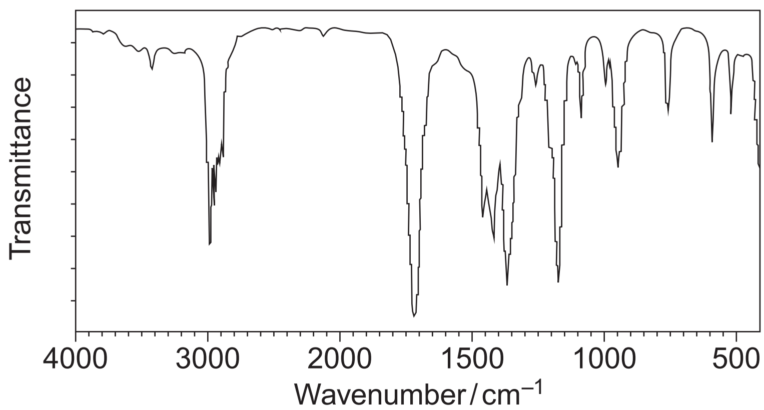
11. (a) Compound **X** contains carbon, hydrogen and oxygen only. It has no reaction with acidified potassium dichromate.

Simplified versions of its mass spectrum, IR spectrum and ^{13}C NMR spectrum are shown.

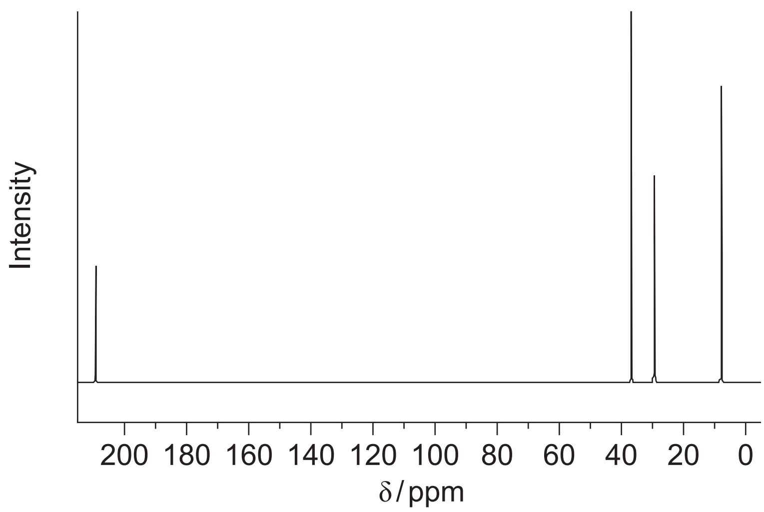
Mass spectrum



IR spectrum



^{13}C NMR spectrum



 (2410U20-1)

Examiner only

Identify compound **X**.

You **must** use information from **all** the sources given and explain how you used it. [8]

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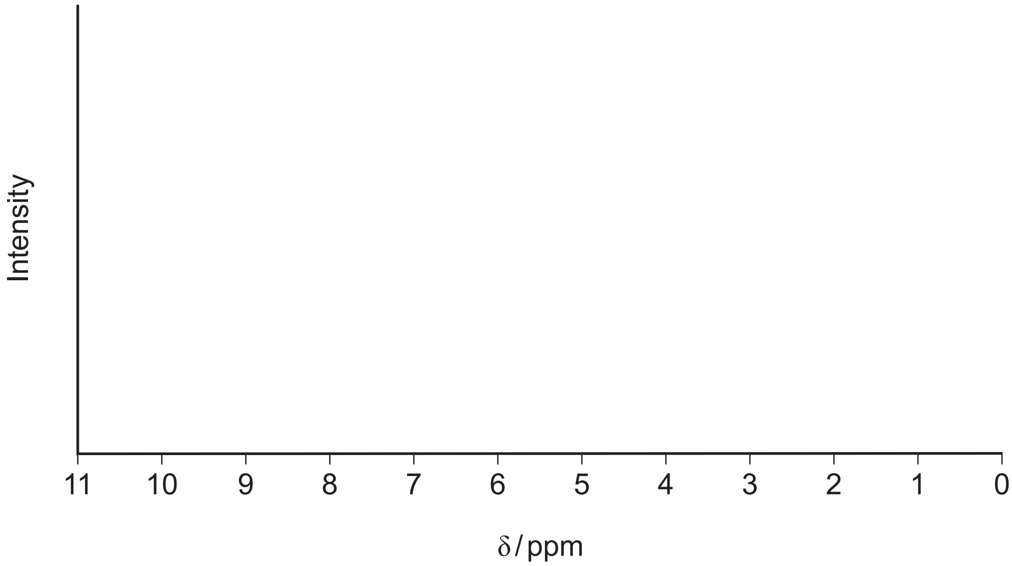
Compound **X**

Examiner only

(b) On the axes below, complete the low resolution ^{1}H NMR spectrum you would expect for the compound you identified in part (a).

You should indicate where you would expect to see peaks **and** the relative intensities of the peaks. [2]



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